OPERATION

The exhaust manifold collects the exhaust gasses exiting the combustion chambers. Then it channels the exhaust gasses to the exhaust pipe attached to the manifold.

REMOVAL

(1) Raise vehicle and disconnect exhaust pipe from the exhaust manifold.
(2) Lower the vehicle.
(3) Disconnect upstream oxygen sensor connector at the rear of exhaust manifold.
(4) Remove the air cleaner bracket (Fig. 93).

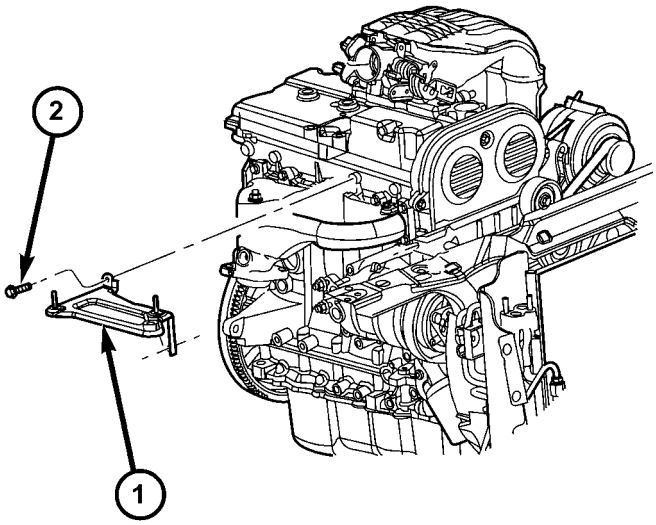

Fig. 93 AIR CLEANER BRACKET

1 - AIR CLEANER BRACKET
2 - BOLT (2)

(5) Remove the heat shield.
(6) Remove the bolts attaching the manifold to the cylinder head.
(7) Remove exhaust manifold.
(8) Inspect the manifold. (Refer to 9 - ENGINE/ MANIFOLDS/EXHAUST MANIFOLD - INSPECTION)

CLEANING

(1) Discard gasket (if equipped) and clean all surfaces of manifold and cylinder head.

INSPECTION

(1) Inspect manifold gasket surfaces for flatness with straight edge. Surface must be flat within 0.15 mm per 300 mm (0.006 in. per foot) of manifold length.
(2) Inspect manifolds for cracks or distortion. Replace manifold as necessary.

INSTALLATION

(1) Clean the manifold mating surfaces.
(2) Install exhaust manifold with a new gasket. Tighten attaching nuts to 20 N·m (175 in. lbs.).
(3) Attach exhaust pipe to exhaust manifold and tighten fasteners to 37 N·m (27 ft. lbs.).
(4) Install and connect the oxygen sensor. (Refer to 14 - FUEL SYSTEM/FUEL INJECTION/O2 SENSOR - COMPONENT LOCATION)
(5) Install the heat shield.
(6) Install the air cleaner bracket.

TIMING BELT COVER(S)

REMOVAL

FRONT COVER

(1) Remove crankshaft vibration damper. (Refer to 9 - ENGINE/ENGINE BLOCK/VIBRATION DAMPER - REMOVAL)
(2) Remove generator drive belt tensioner assembly. (Refer to 7 - COOLING/ACCESSORY DRIVE/ BELT TENSIONERS - REMOVAL)
(3) Remove timing belt front cover bolts, and remove covers.

REAR COVER

(1) Remove front covers.
(2) Remove timing belt. (Refer to 9 - ENGINE/ VALVE TIMING/TIMING BELT/CHAIN AND SPROCKETS - REMOVAL)
(3) Hold camshaft sprocket with Special Tool 6847 while removing center bolt.
(4) Remove timing belt idler pulley.
(5) Remove rear cover fasteners and remove cover from engine.

INSTALLATION

REAR COVER

(1) Install timing belt rear cover and bolts (Fig. 94). Torque bolts to 12 N·m (105 in. lbs).